MATH 110C, HOMEWORK 5

DUE THURSDAY MAY 16

Exercise 1 Let K be a field of positive characteristic p. Let $c \in K$ and let $P = X^p - X - c$. Let L/K be a field extension such that P has a root $\alpha \in L$.

Show that $K(\alpha)$ is a Galois extension of K and that $\mathrm{Gal}(K(\alpha)/K)$ is cyclic.

Exercise 2 Let L/K be a Galois extension with Galois group G and let $\alpha \in L$. Show that $L = K(\alpha)$ if and only if $g(\alpha) \neq \alpha$ for all non-trivial elements $g \in G$.

Exercise 3 Let L/K be a finite algebraic extension of characteristic 0 fields. Let $\{e_1, \ldots, e_n\}$ be a basis of L over K. Let H be a subgroup of $\mathrm{Gal}(L/K)$.

Given $i \in \{1, \ldots, n\}$, let $f_i = \sum_{h \in H} h(e_i)$.

Show that $L^H = K(f_1, \ldots, f_n)$.